YASKAWA

DCI JOB EXCHANGER
COMMUNICATION MANUAL

Advanced Robot Controller Communication Service

Upon receipt of the product and prior to initial operation, read these instructions thoroughly, and retain for future reference.

Have the following information available when contacting the YASKAWA Representative:

- System
- Primary Application
- Software Version *(Located on Programming Pendant by selecting: {Main Menu} - {System Info} - {Version})*
- Warranty ID *(Located on Robot Controller)*
- Robot Serial Number *(Located on Manipulator data plate)*
- Robot Sales Order Number *(Located on Robot controller data plate)*

The Americas YASKAWA Representative

- 24-hour Telephone Number: **(937) 847-3200**
  Use for urgent or emergency needs for technical support, service and/or replacement parts
- Routine Technical Inquiries: *techsupport@motoman.com*
  Allow up to 36 hours for response

MANUAL NO.
197138-1CD ⓪

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Safety

For Your Safety

Robots generally have requirements which are different from other manufacturing equipment, such as larger working areas, high-speed operation, rapid arm movements, etc., which can pose safety hazards.

Read and understand the instruction manuals and related documents, and observe all precautions in order to avoid the risk of injury to personnel and damage to equipment.

Carelessness contributes to serious accidents in the work area.

It is the user's responsibility to ensure that all local, state, and national codes, regulations, rules, or laws relating to safety and safe operating conditions are met and followed.

DANGER

- Teaching, operations, and maintenance of the Robot must conform to:
  - Industrial Safety and Health Law
  - Order for Enforcement of the Industrial Safety and Health Law
  - Industrial Safety and Health Regulations
  - Technical Standards for Electrical Facilities

  Other related laws and regulations are:
  - Occupational Safety and Health Act in USA
  - Factory Act (Gewerbeordnung) in Germany
  - Health and Safety at Work, etc. Act in UK
  - EC Machinery Directive 2006/42/EC
- Prepare:
  - SAFETY WORK REGULATIONS
    based on concrete policies for safety management complying with related laws and regulations.
- Observe:
  - JIS B 8433-1: 2015 "Robots for industrial environments-Safety requirements" (ISO 10218-1: 2011) for safe operation of the robot. (JIS B 8433 is for Japan only)
- Reinforce:
  - SAFETY MANAGEMENT SYSTEM
    by designating authorized operators and safety managers for the Robot, as well as giving continuing safety education and training.
- Teaching, operation, and maintenance of the Robot are specified as "Hazardous Operations" in the Industrial Safety and Health Act (for Japan only).
  Personnel engaged in these operations must receive special training offered by YASKAWA.

DANGER

- Personnel engaged in operation, maintenance, or management of the Robot must receive required training before using the Robot.

For more information on training, contact a YASKAWA representative.

- Make sure to have and follow all manuals, read them thoroughly and understand the contents of them.

Confirm that you have all required manuals. If any of the manuals are missing, contact a YASKAWA representative.

- Read “Safety” of the Controller instructions before operating.

Not reading and understanding chapter 1 of the Controller instruction can result in death or serious injury.

We suggest that you obtain and review a copy of the ANSI/RIA National Safety Standard for Industrial Robots and Robot Systems (ANSI/RIA R15.06-2012). You can obtain this document from the Robotic Industries Association (RIA) at the following address:

Robotic Industries Association
900 Victors Way
P.O. Box 3724
Ann Arbor, Michigan 48106
TEL: (734) 994-6088
FAX: (734) 994-3338
www.roboticsonline.com

Ultimately, well-trained personnel are the best safeguard against accidents and damage that can result from improper operation of the equipment. The customer is responsible for providing adequately trained personnel to operate, program, and maintain the equipment.

We recommend approved YASKAWA training courses for all personnel involved with the operation, programming, or repair of the equipment.

This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications.

WARNING

- Safe operation of this equipment is the users responsibility.
  - The conditions under which the equipment will be operated safely should be reviewed by the user. The user must be aware of the various national codes, ANSI/RIA R15.06-2012 safety standards, and other local codes that may pertain to the installation and use of this equipment.

Not following all national codes, safety standards and local codes can result in death or serious injury

CAUTION

- Only trained personnel familiar with the operation, manuals, electrical design, and interconnections of this equipment should program, or maintain the system.

Any personnel involved with the operation of the equipment must understand potential dangers of operation.

NOTICE

- The drawings and photos in this manual are examples. Differences may exist between them and the delivered product.
- YASKAWA may modify this model without notice due to product improvements, modifications, or changes in specifications. If such modification is made, the manual number will also be revised.
- Some operations require standard passwords and while others require special passwords.
- If a manual is damaged or lost, contact a YASKAWA representative to order a new copy. Make sure to tell the YASKAWA representative the Manual Number listed on the front cover.

Notes for Safe Operation

Read this manual carefully before installing, operating, maintaining, or inspecting the system.

In this instruction, Safe Operations are classified as “DANGER”, “WARNING”, “CAUTION” or “NOTICE”.

DANGER	Indicates an imminently hazardous situation which, if not avoided, **WILL result in death or serious injury**.
WARNING	Indicates a potentially hazardous situation which, if not avoided, **MAY result in death or serious injury**.
CAUTION	Indicates a hazardous situation, which if not avoided, **MAY result in minor to moderate injury**.
CAUTION	Indicate a situation which if not avoided **may result in equipment damage.**
NOTICE	Indicates **practices not related to personal injury**.

NOTICE
To ensure safe and efficient operation at all times, be sure to follow all instructions, even if not designated as “DANGER”, “WARNING” or “CAUTION”.

Definition of Terms Used Often in This Manual

The Robot is the YASKAWA industrial robot product.

The Robot usually consists of a Robot, Controller, Programming Pendant, and Robot cables.

In this manual, the equipment is designated as follows:

Equipment	Manual Designation
YRC1000micro, YRC1000, DX200, FS100, DX100, or NX100 controller	Robot Controller
YRC1000micro, YRC1000, DX200, FS100, DX100, or NX100 Programming Pendant	Programming Pendant
Manipulator	Robot
Cable between the Robot and the Controller	Robot cable
Positioner	Positioner

Descriptions of the Programming Pendant keys, buttons, and displays are shown as follows:

Equipment		Manual Designation
Programming Pendant	Character Keys /Symbol Keys	The keys which have characters or symbols printed on them are denoted with [ ]. e.g. [ENTER]
	Axis Keys /Numeric Keys	[Axis Key] and [Numeric Key] are generic names for the keys for axis operation and number input.
	Keys pressed simultaneously	When two keys are to be pressed simultaneously, the keys are shown with a “+” sign between them, e.g. [SHIFT]+[COORD].
	Mode Switch	Mode Switch can select three kinds of modes that are denoted as follows: REMOTE, PLAY or TEACH. (The switch names are denoted as symbols)
	Button	The three buttons on the upper side of the Programming Pendant are denoted as follows: START, HOLD, or EMERGENCY STOP. (The button names are denoted as symbols)
	Displays	The menu displayed in the Programming Pendant is denoted with { }. e.g. {JOB}

Registered Trademark

In this manual, names of companies, corporations, or products are trademarks, registered trademarks, or bland names for each company or corporation. The indications of ® and ™ are omitted.

YASKAWA Representative Information

The following information is for a YASKAWA representative in America, for other locations refer to the back cover.

(937) 847-3200

For **routine** technical inquiries, you can also contact a YASKAWA representative at the following e-mail address:

techsupport@motoman.com

When using e-mail to contact a YASKAWA representative, please provide a detailed description of your issue, along with complete contact information. Please allow approximately 24 to 36 hours for a response to your inquiry.

NOTICE

Please use e-mail for routine inquiries only. If you have an urgent or emergency need for service, replacement parts, or information, you must contact a YASKAWA representative at the telephone number shown above or on the back cover.

Please have the following information ready before calling a YASKAWA representative:

- Software — DCI Job Exchanger
- Manipulator _____
- Positioner _____
- Primary Application _____
- Robot Controller _____
  YRC1000micro, YRC1000, DX200, FS100, DX100, or NX100
- Software Version _____
  Access this information on the Programming Pendant's LCD display screen by selecting {MAIN MENU} - {SYSTEM INFO} - {VERSION}
- Robot Serial Number _____
  Located on the Manipulator data plate
- Robot Sales Order Number _____
  Located on the Robot Controller data plate
- Warranty ID: _____
  Located on Robot Controller

Table of Contents

1 Introduction and Requirements 11
1.1 Introduction 11
1.2 System Requirements 11
1.3 Reference Documentation 12
2 Setup 13
2.1 PC Installation 13
2.2 Firewall Access 13
2.3 Robot Setup 13
2.3.1 For YRC1000 or Newer Controllers 13
2.3.2 For Older Generation Controllers 14
3 Operation 15
3.1 First Launch 15
3.2 Main Window 16
4 INFORM DCI Commands 18
4.1 LOADJ 18
4.2 SAVEJ 18
4.3 DELETEJ 19
4.4 SWAIT 19
5 Alarm Codes 20

1 Introduction and Requirements

1.1 Introduction

Data Communication by Instruction (DCI) Job Exchanger is an advanced robot controller communication service. It serves requests from the robot's INFORM program to load and save job files. DCI Job Exchanger also supports simultaneous communication with multiple robot controllers.

1.2 System Requirements

– PC with Windows 10 or newer.

– Robot Controllers:

- YRC1000micro
- YRC1000
- DX200
- FS100
- DX100
- NX100

– Ethernet function enabled on robot controller.
(NX100 requires PCI add-on board.)

NOTICE

THE DCI Job Exchanger is **NOT** compatible with Host Control function (eg: MotoCom, Ethernet Server, Smart Pendant, Cockpit).

1.3 Reference Documentation

For additional information concerning the individual components, refer to the following documentation. The following table include links that allow you to click on the link and it will take you to the referenced manual..

Manuals	Manual Number
Robot Controller Manuals	
• *YRC1000micro Controller Manual*	*RE-CTO-A222*
• *YRC1000 Controller Manual*	*RE-CTO-A221*
• *DX200 Controller Manual*	*RE-CTO-A220*
• *FS100 Controller Manual*	*RE-CTO-A218*
• *DX100 Controller Manual*	*RE-CTO-A215*
• *NX100 Controller Manual*	*RE-CTO-A211*
Robot Controller INFORM User's Manual	
• *YRC1000micro INFORM User's Manual*	*RE-CKI-A468*
• *YRC1000 INFORM User's Manual*	*RE-CKI-A466*
• *DX200 INFORM User's Manual*	*RE-CKI-A464*
• *FS100 INFORM User's Manual*	*RE-CKI-A458*
• *DX100 INFORM User's Manual*	*RE-CKI-A452*
• *NX100 INFORM User's Manual*	*RE-CKI-A444*
• *NX100 PCI Board*	*154309-1CD*
Vendor manuals for system components not manufactured by YASKAWA	

2 Setup

2.1 PC Installation

NOTICE

- Before installing the DCI Job Exchanger software, remove the Sentinel Security USB dongle from the computer, if required.
- While running the setup execution file it will also install a Sentinel Security Driver. Make sure to “Complete” installation of this driver.

1. Insert the DCI Job Exchanger software provided on a USB flash drive into the USB port on the computer.
2. Run the setup.exe file.
3. Install the Sentinel Security USB dongle into the USB port on the computer.

2.2 Firewall Access

1. Make exceptions in any network firewall to allow communication over ports UDP 10000 and UDP 10006.
2. Allow passage of ICMP (Ping) traffic in the firewall.

2.3 Robot Setup

A single instance of the DCI Job Exchanger application can be used to communicate with multiple robot controllers. But each robot controller must be individually configured to support DCI communication.

Contact your local YASKAWA Support Representative for assistance with enabling the Ethernet function.

The following robot parameters are required to use DCI communications. These can be set on the programming pendant with MANAGEMENT security level or higher.

- RS000 = 2: Ethernet Protocol BSC-like
- RS005 = 0: Disable Host Control
- RS007 = 0: Prohibit Writing with Host Control
- RS029 = 1: Allow loading data in Play mode
- RS034 = 75: Response wait timer (0.1 seconds)
- RS039 = 255: Alarm generation method (continue indefinitely)

Additionally, ensure that Pseudo Input #82015 (CMD REMOTE SEL) is disabled.

2.3.1 For YRC1000 or Newer Controllers

1. Select [Ex Memory] ⟶ [Comm Setting (Expand)] from the Main Menu on the Programming Pendant.
2. Set the [Host Address (PC)] to the IP address of the PC that is running DCI Job Exchanger.

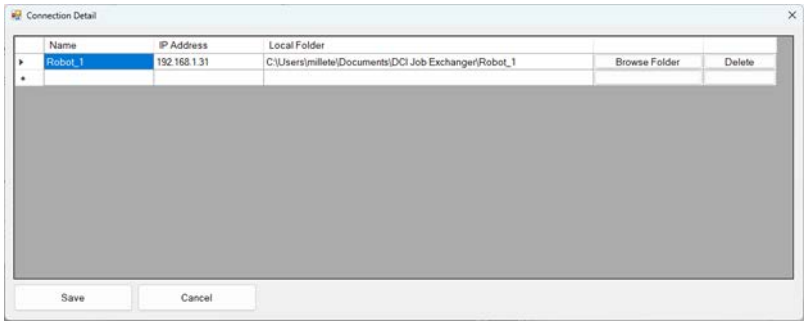
2.3.2 For Older Generation Controllers

Enter the IP address of the PC in robot parameters RS082 - RS085. For example, if the IP address of the PC is 192.168.1.100, then RS082 = 192, RS083 = 168, RS084 = 1, RS085 = 100.

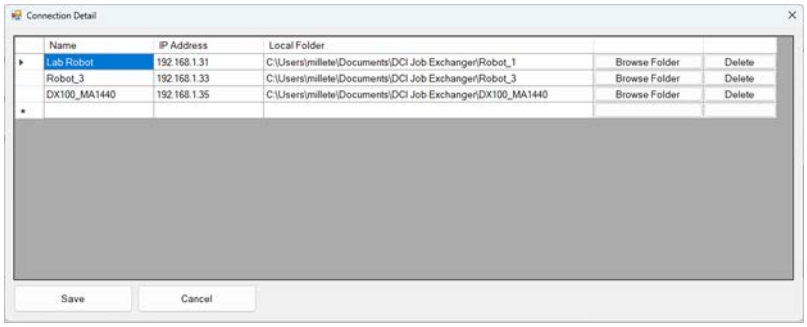
3 Operation

3.1 First Launch

The first time opening the DCI Job Exchanger, it will have a prompt with the Connection Detail screen.



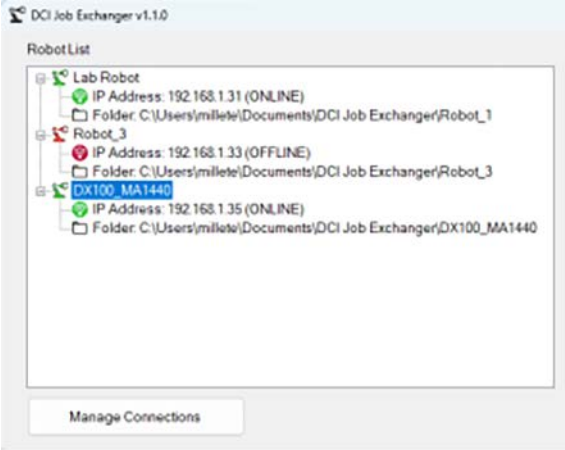
1. Provide the following for the robot controller:
   - Name
   - IP Address
   - Local Folder - the path to a folder on the PC which will store the jobs that will be transferred.



2. Repeat *step 1* for all robots to be added.
3. Press [Save] to register the settings and navigate to the Main Window.

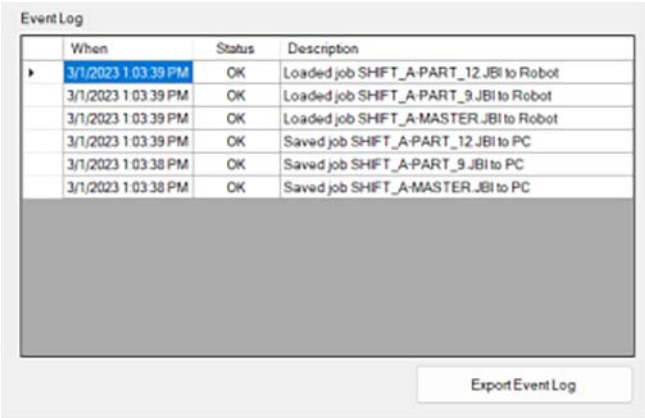
3.2 Main Window

The main window shows the status of all robot controllers that were configured in the Connection Detail screen.



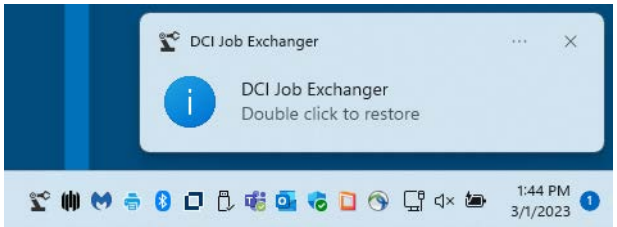
– Icons for each robot indicate the connection status by changing color.
  - **GREEN - (ONLINE)** or
  - **RED (OFFLINE)**.

– Selecting the robot icon opens the Event Log that shows any communication activity that has occurred since launching the DCI Job Exchanger.



– Double-clicking on a folder under each robot opens Windows Explorer to the directory where job files are stored.

– Minimizing the DCI Job Exchanger window will cause it to hide the window while continuing to run in the background. The logo appears in the system tray. Double-click to restore the main window.



As long as the DCI Job Exchanger application stays open or minimized, it will continue to listen for DCI commands from each of the robot controllers in this list.

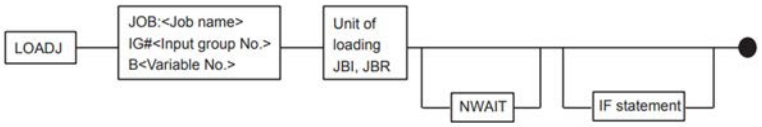
4 INFORM DCI Commands

NOTICE

The DCI Job Exchanger software does not support LOADV and SAVEV commands.

4.1 LOADJ

The LOADJ command will retrieve the specified job file from the PC and load it into the robot controller's memory.



If the robot controller's memory contains a job with the same name as the job installing, the new job overwrites the existing job. However, an alarm occurs if the job loading is currently under execution or registered in the current call stack.

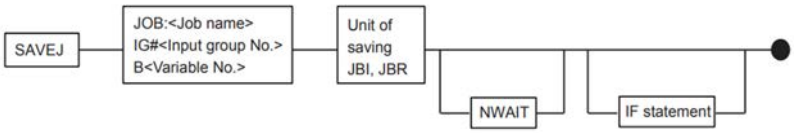
If the target job specifies using and input group (IG#) or variable (B), it performs in the same way as the CALL command. If the current value is 0, then the transfer will not perform.

When the NWAIT tag is specified, the next instruction is executed without waiting for completion of the file transfer. An alarm will occur if calling a specified job before completing transfer.

If another LOADJ or SAVEJ operation is already in process, then the system will wait for that to complete, regardless of the NWAIT tag.

4.2 SAVEJ

The SAVEJ command sends a copy of a specified job file to the PC for storage.



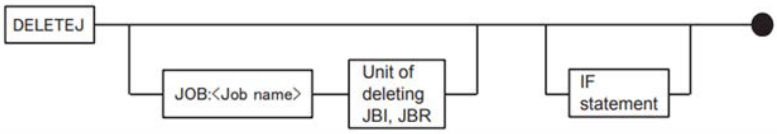
If the target job specifies using and input group (IG#) or variable (B), it performs in the same way as the CALL command. If the current value is 0, then the transfer will not perform.

When the NWAIT tag is specified, the next instruction is executed without waiting for completion of the file transfer. An alarm will occur if calling a specified job before completing transfer.

If another LOADJ or SAVEJ operation is already in process, then the system will wait for that to complete, regardless of the NWAIT tag.

4.3 DELETEJ

The DELETEJ command deletes all jobs except its own from the robot controller's memory. This command can be set to delete a specific job.



Jobs do not delete if they are under execution or registered in the current call stack.

4.4 SWAIT

Waits for completion of loading or saving jobs. Use this command to recognize the completion of LOADJ or SAVEJ command when specifying a NWAIT for these instructions.

5 Alarm Codes

Code	Message	Subcode
4104	WRONG EXECUTION OF LOAD INST	Refer to table below
4105	WRONG EXECUTION OF SAVE INST	
4106	WRONG EXECUTION OF DELETE INST	

Sub-code	Meaning
1	Insufficient memory capacity
2	Job editing prohibited
3	Attempted to load or delete a job being executed
4	No specified job
17	Instruction destroyed
19	Invalid character in job name
20	Invalid character in label
23	Invalid character in this system
24	Syntax error
90	Control command sending/receiving error (Ethernet)
104	Error response from host computer
111	Syntax error
112	Error in position data
113	No NOP or END instruction
117	Format error
118	Invalid number of data
120	Data range exceeded
122	Destroyed file exists
127	This protocol already used.
128	File accessing in other function
211	System block error (Receiving EOT while waiting ACK)
212	System block error (Receiving EOT at start receiving)
213	System block error (Receiving EOT before receiving the last block)
214	System block error (Receiving codes other than EOT before receiving the last block)
221	Sending error (Retry for NAK exceeded)
222	Sending error (Timeout for timer A retry)
223	Sending error (ACK0/ACK1 order error after retry)
231	Receiving error (Timeout for timer A while waiting ACK after ENQ, timeout for timer A while waiting ENQ response)
232	Receiving error (Timeout for timer B while receiving a text)
233	Receiving error (Heading length is shorter than 6 characters)
234	Receiving error (Heading length is longer than 6 characters)
235	Receiving error (Header number error)
236	Receiving error (Text length exceeds 256 bytes)

Sub-code	Meaning
237	Receiving error (Receiving other than ENQ while waiting ENQ, receiving other than control code while waiting control code, receiving other than STX, SOH, ENQ, EOT while waiting text)
240	Software error
241	Hardware error (Overrun)
242	Hardware error (Parity error)
245	Hardware error (Framing error)
244	Hardware error (Sending timeout (timer A))
245	Hardware error (Sending timeout (timer B))

DCI JOB EXCHANGER
COMMUNICATION MANUAL

Advanced Robot Controller Communication Service

For inquiries or after-sales service on this product, contact your local YASKAWA representative as shown below.

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Specifications are subject to change without notice
for ongoing product modifications and improvements.

YASKAWA

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